Republic of the Philippines

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LAB REPORT I

(ARRAY APPLICATION [1D])

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DATE:

NOVEMBER 9,2024

MACHINE PROBLEM DESCRIPTION:

Finding a Defective Machine in a Factory

This Python program is a simulation of a factory where a number of machines will be producing products. Each machine has an ID, and as each one is supposed to have appeared twice in the dataset, because it produces an even number of items, but a machine has gone **faulty**, which, on producing items, will result in an odd number of items being produced, so its ID will appear just once in the list. The overall aim of the program is to identify the faulty machine in both the 1D array and the 2D array.

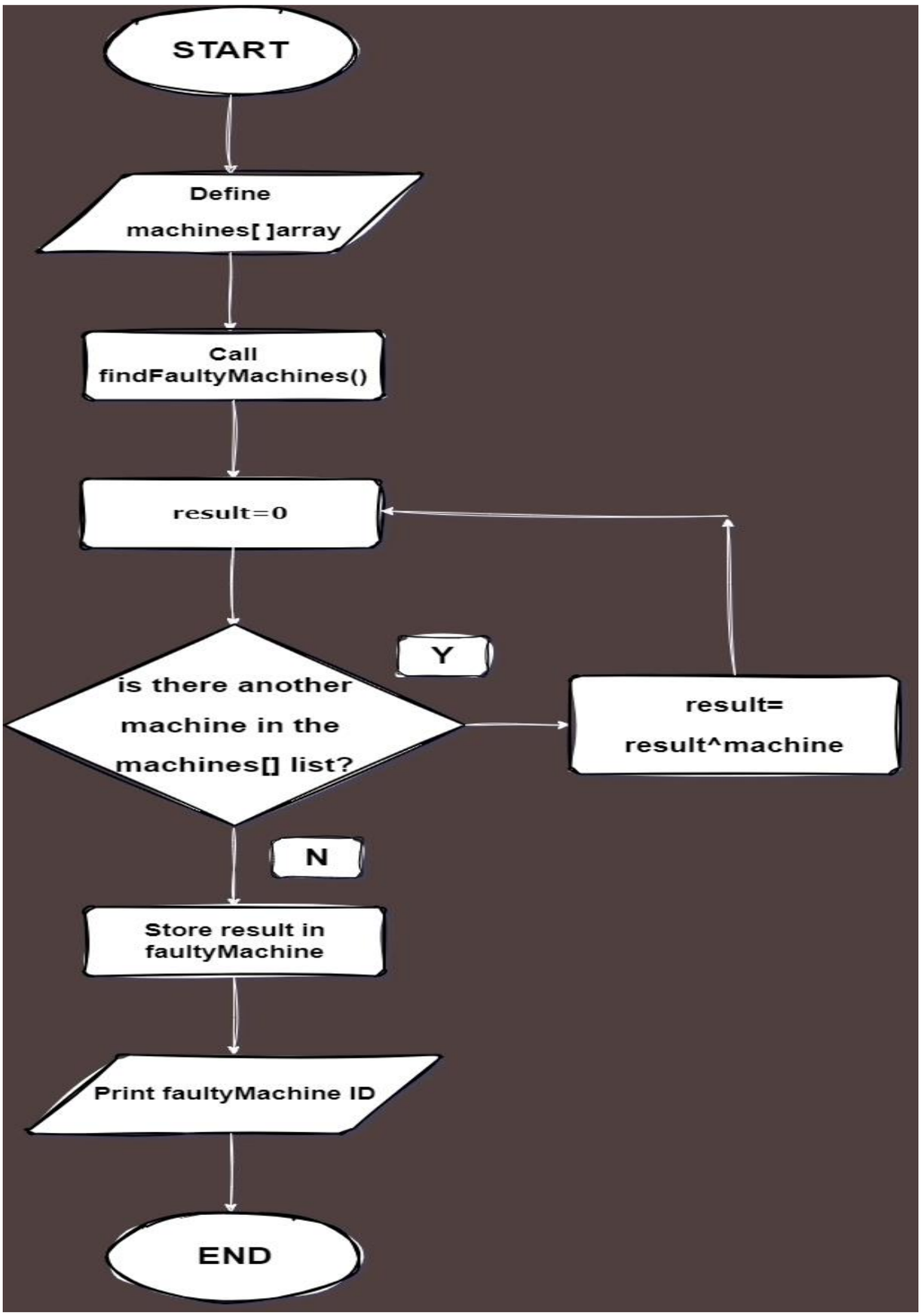
1D Array:

In the 1D version, the program runs merely on a single set of machine IDs. The XOR operation is then performed on the list of machine IDs such that all the duplicates cancel out, and only the ID of the faulty machine remains there.

FLOWCHART:

1D Array Flowchart

This flowchart represents a scenario where the program processes a single set of machine IDs (1D array) and uses the XOR operation to identify the faulty machine.



SOURCE CODE:

1D Array Code

The code processes a single set of machine IDs using the XOR operation to find the faulty machine**:**

findTheFaultyMachine_1D.py

```python

def findFaultyMachine(machines):
    result = 0
    for machine in machines:
        result = result ^ machine
    return result

def main():
    machines = [101, 202, 303, 404, 101, 303, 404]
    faultyMachine = findFaultyMachine(machines)

    print(f"The faulty machine is: {faultyMachine}")
if __name__ == "__main__":
    main()
```

SAMPLE OUTPUT:

1D Array Sample Output:

Here, the program has identified 202 as the faulty machine, since all other ID's occur twice.

```
Run    findTheFaultyMachine_1D
"C:\Users\Gian Carlo Trilles\PycharmProjects\pythonProject
The faulty machine is: 202

Process finished with exit code 0
```